



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part – 2 (Paper-3)

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Functions in C++

- A function is a group of statements that together perform a task. Every program in C++ has **main()** function by default. C++ standard library provides lot of built in functions that you can call or use in your program.

OR

- **A function is a block of code that performs a specific task and can be reuse anywhere in the program.**
- All function Parentheses () are present after the name
Example: add(), hira(), main(),.....
- Syntax :

```
return_type function_name(parameters)
{
    // function body
}
```

- Example :

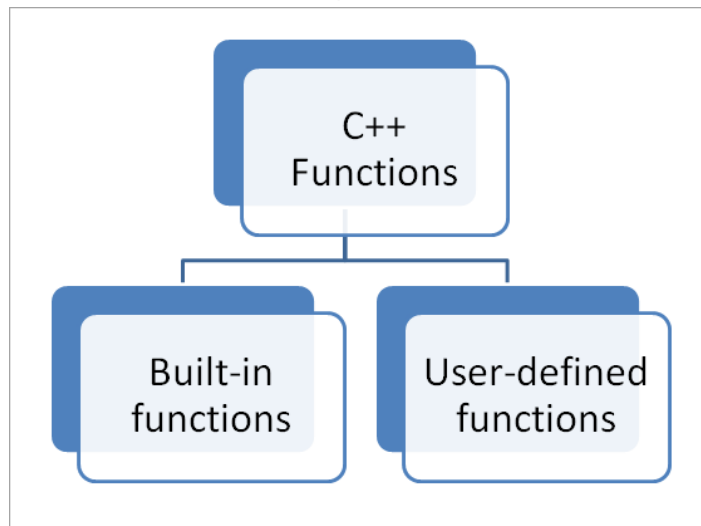
```
int add(int a, int b)
{
    return a + b;
}

int main()
{
    cout << add(10, 20);
}
```

Output: 30

There are two types of function:

1. Standard Library Functions / Predefined Function / Built in Functions
2. User-defined Function / Custom Function



1. Predefine Function

- Predefine functions are the functions that are already written and stored in c++ standard libraries. Programmers can directly use them in their program without writing their definition.
- You only need to include the correct header file to use them.

OR

- A predefined function in C++ is a function that is already define in the C++ standard library and can be used directly by including the appropriate. header file.

• **Syntax**

```
#include <header_file>  
function_name(arguments);
```

Example

```
#include <cmath>  
int x =sqrt(25);
```

❖ **main() function**

- The main() function which is the starting point for the execution of a program. In C++, the main() returns a value of type int to the operating system. The functions that have a return value should use the return statement for termination. The main() function in C++ is, therefore, defined as follow :

```

• int main()
{
    .....
    .....
    Return 0;
}

```

Most C++ compilers will generate an error or warning if there is no return statement.

2. User define Function

- A User defined function in C++ is a function that is define by the programmer to perform a specific task and can be called whenever needed in the program.

- Syntax

```

void hira()
{
    // code to be executed
}

```

Here, **hira()** is the name of the function.

void means that the function does not have a return value.

inside the function (the body), **add code** that defines what the function should do.

❖ List of Predefine / Library Function in header files in c++ and its use/work

1. <iostream>

Used for input and output operations in C++.

Name	Use	Type
cout	Display normal output on the screen(Standard output)	ostream object
cin	Takes input from the keyboard without space (standard input)	istream object
getline()	Reads an entire line, including spaces	Function

endl	Inserts a new line and flushes the output buffer	Manipulator
ws	Consumes leading whitespace from a input stream	manipulator
flush	Flushes the output buffer	manipulator

➤ **Manipulator** : A manipulator is a special facility used to control / modify the formatting or behavior of an input/output stream. Manipulator is normally used with an I/O stream, such as cout or cin.

➤ **Buffer memory** : Buffer memory is a temporary storage area that holds data until it is ready to be processed or transferred.

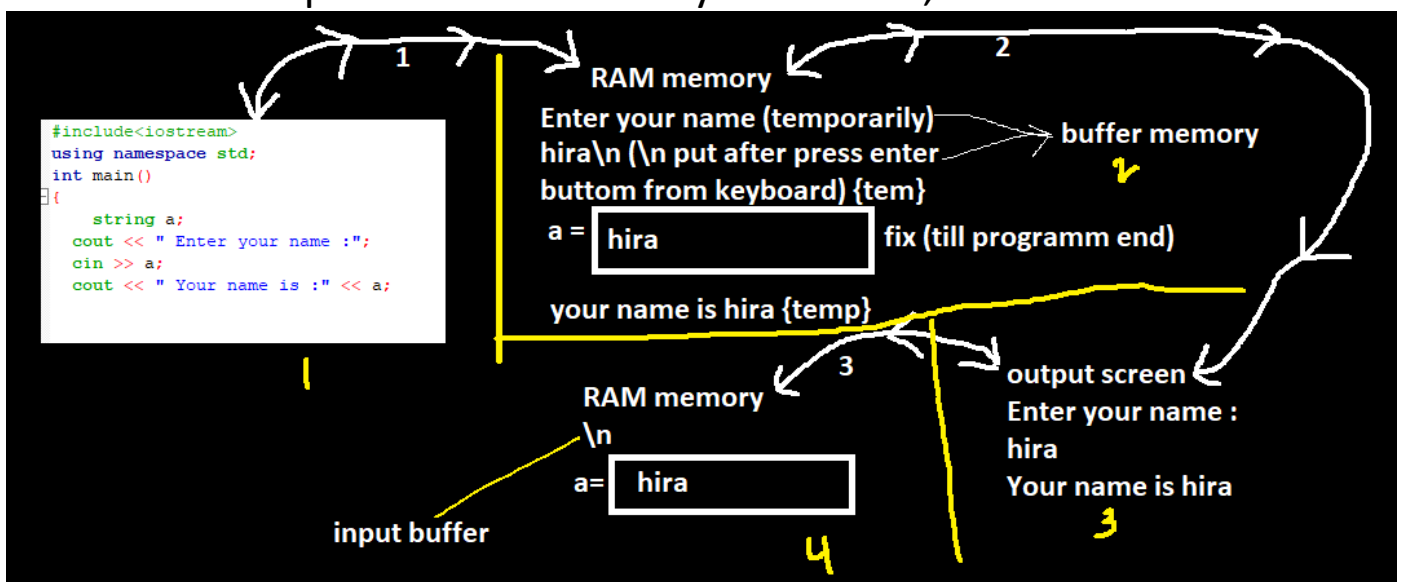
Two types of buffer memory

(i) **Input buffer** : Temporarily stores data received from an input device, such as the keyboard before the program reads it.

Example : cin << a;

(ii) **Output buffer** : Temporarily stores output data before it is sent to the screen or another output device.

Example : cout << "enter your name";



Note : मान लीजिए आप लिखते हैं:

hira kumar[Enter]

Keyboard से input stream (stream means data flow process/channel) में लगभग यह sequence आता है:

h i r a k u m a r \n

यहाँ:

- hira → characters
- space → space character
- kumar → characters
- Enter → \n (newline)

इसलिए जब आप: cin >> s; करते हैं, तो cin केवल पहला word पढ़ता है:

s = "hira"

लेकिन बाकी input buffer में रहता है:

HIRA KUMAR

Ekumar\n

फिर अगर: cin.ignore(); लगाते हैं, तो वह सिर्फ पहला character (space) हटाता है: kumar\n रह जाता है |

और:

getline(cin, city); → kumar पढ़ लेता है।

अगर आप चाहते हैं कि input buffer में मौजूद पूरा बाकी all input clear हो जाए, तो cin.ignore() को arguments के साथ use कर सकते हैं:

```
cin.ignore(1000, '\n');
```

इसका मतलब:

अधिकतम 1000 characters ignore करो, या जब \n मिल जाए तो रुक जाओ।

या Better/general form

```
cin.ignore(numeric_limits<streamsize>::max(), '\n');
```

इसके लिए: #include <limits> header file का use करना चाहिए।

➤ Flush : Send the buffered data (buffer memory data) to its destination immediately (output screen,....)

```
hello.cpp X
#include<iostream>
using namespace std;
int main()
{
    string s,city;
    cout << " Welcome C++ programming language";
    cout << endl;
    cout << "what is your name :";
    cin >> s;
    cout <<" your name is : " << s << flush;
    cout <<flush;
    cout << endl << "what is your city name : ";
    cin.ignore( 1000, '\n'); // ignore \n
    getline (cin, city);
    cout << "your city name is : " << city;
}
```

C:\Users\hira\Desktop\hello.exe
Welcome C++ programming language
what is your name :hira kumar
your name is : hira
what is your city name : bihar sharif
your city name is : bihar sharif
Process returned 0 (0x0) execution time : 5.988 s
Press any key to continue.

2. <cmath> or <math.h>

<cmath> use modern c++ preferred.<math.h> old c++ also support.

<cmath> is a standard c++ header file that provides mathematical functions for performing calcuations.

Basic mathematical functions		
Function name	Use	Example
abs()	Absolute value	abs(-10) = 10
fabs()	Absolute value of floating number	fabs(-10.5)=10.5
fmod()	Floating point remainder	fmod(10.5,3)=1.5
fmax()	Returns larger floating/int value	fmax(10.5,20.5)=20.5
fmin()	Returns smaller floating/int value	fmin(10.5,20.5)=10.5

```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
cout << "abs(-10 \t =" << abs(-10) << endl;
cout << "fabs(-10.5)\t =" << fabs(-10.5) << endl;
cout << "fmod (10.5,3)\t =" << fmod(10.5,3) << endl;
cout << "fmax(10.5,20.5)\t =" << fmax(10.5,20.5) << endl;
cout << "fmin(10.5,20.5)\t =" << fmin(10.5,20.5) << endl;
cout << "fmin(58,45)\t =" << fmin(58,45) << endl;
cout << "smaller value of 25,98,46 is : " << fmin(25,(fmin(98,46))) << endl;
}
```

C:\Users\hira\Desktop\hello.exe

```
abs(-10)           =10
fabs(-10.5)        = 10.5
fmod (10.5,3)       =1.5
fmax(10.5,20.5)     =20.5
fmin(10.5,20.5)     =10.5
fmin(58,45)         =45
smaller value of 25,98,46 is : 25
```

Questions

1. Write a C++ program to find the absolute value of -125 using abs().
2. Write a C++ program to find the absolute value of -45.75 using fabs().
3. Write a C++ program to find the remainder when 25.5 is divided by 4.2 using fmod().
4. Write a C++ program to find the largest value among 25, 98, and 46 using fmax().
5. Write a C++ program to find the smallest value among -15.5, 20.75, and -8.25 using fmin().
6. Write a C++ program to find the difference between the largest and smallest values among -25.5, 40.75, -10.25, and 60.5.
7. Write a C++ program to find the remainder of the difference between -85.5 and 32.25 when divided by 7.5.
8. Write a C++ program to find the smallest absolute value among -45, 28, -12, and 35.
9. Write a C++ program to find the largest absolute value among -125.75, 98.5, -150.25, and 75.25.
10. Write a C++ program to find the remainder of the largest value divided by the smallest value among -18.5, 25.75, -32.25, and 12.5.

Answer : 125,45.75,0.3,98,-15.5,86.25,5.25,12,150.25,25.75

Power and root

Function name	Use	Example
sqrt()	Square root	sqrt(25) = 5
cbrt()	Cube root	cbrt(27) = 3
pow()	Raises a number to a power	pow(2,3)=8
hypot()	Calculates $\sqrt{a^2 + b^2}$	hypot (3,4)=5

```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
cout << "square of 5 is \t = " << 5*5 << endl;
cout << "square root of 25 is \t = " << sqrt(25) << endl;
cout << "cube root of 27 is \t = " << cbrt(27) << endl;
cout << "2 power 3 \t\t = " << pow(2,3) << endl;
cout << "square root of (3 square + 4 square) \t = " << hypot(3,4);
}
```

C:\Users\hira\Desktop\hello.exe

```
square of 5 is           = 25
square root of 25 is     = 5
cube root of 27 is       = 3
2 power 3                = 8
square root of (3 square + 4 square) = 5
```

Rounding functions

Function name	Use	Example
ceil()	Rounds upward	ceil(4.2) = 5
floor()	Rounds downward	floor(4.8) = 4
round()	Rounds to nearest integer	round(4.6) = 5
trunc()	Removes fractional part	trunc(4.8) = 4
rint()	Rounds a/c to current rounding mode (nearest even when .5) {otherwise same round() function}	rint(4.8) = 4

```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
cout << "ceil(4.2) \t = " << ceil(4.2) << endl;
cout << "floor(4.8) \t = " << floor(4.8) << endl;
cout << "round(4.6) \t = " << round(4.6) << endl;
cout << "round(4.5) \t = " << round(4.5) << endl;
cout << "round(4.1) \t = " << round(4.1) << endl;
cout << "rint(4.6) \t = " << rint(4.6) << endl;
cout << "rint(5) \t = " << rint(5) << endl;
cout << "rint(4.1) \t = " << rint(4.1) << endl;
cout << "rint(4.5) \t = " << rint(4.5) << endl;
}
```

C:\Users\hira\Desktop\hello.exe

```
ceil(4.2)           = 5
floor(4.8)          = 4
round(4.6)          = 5
round(4.5)          = 5
round(4.1)          = 4
rint(4.6)           = 5
rint(5)             = 5
rint(4.1)           = 4
rint(4.5)           = 4
```

Questions

- Write a C++ program to calculate 2^8 and round the result to the nearest integer.
- Write a C++ program to calculate the square root of 245.75 and round the result to the nearest integer.
- Write a C++ program to calculate $5^{3.5}$ and round the result to the nearest integer.
- Write a C++ program to calculate the cube root of 125.75 and round the result to two different integer values using floor() and ceil().
- Write a C++ program to calculate the square root of 3^8 and display the result using round().

16. Write a C++ program to calculate $12^{2.5}$, find its square root, and round the final result to the nearest integer.
17. Write a C++ program to calculate the fifth root of 2500 using a power function and round the result to the nearest integer.
18. Write a C++ program to calculate $\sqrt{1250.5}$, then display its rounded-down, rounded-up, and nearest integer values.
19. Write a C++ program to calculate $7^{4.2}$, find its square root, and display the final result using rint().
20. Write a C++ program to calculate the cube root of 9876.54, square the result, and display the final value using floor(), ceil(), and round().

Answer : 256,16,280,(5, 6),81,49,5,(35, 36, 35),98,(98, 99, 98)

Trigonometric functions

Function name	Use	Example
sin()	Sine	sin(0) = 0
cos()	Cosine	cos(0)=1
tan()	Tangent	tan(0) =0

Note : Sine (x will be insert in radian value not degree)

$$\text{Radian} = \text{degree} \times \frac{\pi}{180} \quad (\pi = 3.141)$$

```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
//cout <<"value of sin(0) = " << sin(0) << endl; (no, it is wrong)
cout << " value of sin(0) = " << sin(0*3.14/180) << endl;
cout << " value of cos(0) = " << cos(0*3.14/180)<< endl;
cout << " value of tan(0) = " << tan(0*3.14/180)<< endl;
cout << " value of cosec(0) = " << 1/sin(0*3.14/180)<< endl;
}
```

C:\Users\hira\Desktop\hello.exe

```
value of sin(0) = 0
value of cos(0) = 1
value of tan(0) = 0
value of cosec(0) = inf
```

Other functions

Function name	Use	Example
Exp()	Calculates e^x	Exp(1) = 2.71828
Log()	Natural logarithm	Log(1) = 0
Log10()	Base-10 logarithm	Log10(100) =2
Isgreater()	Checks > (output 0,1)	Isgreater(10,5)=1

Isless()	Checks <(output 0,1)	Isless(10,5) =0
Isgreaterequal()	Checks>=	Isgreaterequal(10,10)=1

```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
cout << " exp(1) = " << exp(1) << endl;
cout << "log(1) = " << log(1) << endl;
cout << "log10(100) = " << log10(100) << endl;
cout << "isgreater(10,5) = " << isgreater(10,5) << endl;
cout << "isless(10,5) = " << isless(10,5) << endl;
cout << "isgreaterequal = " << isgreaterequal(10,10) << endl;
```

```
C:\Users\hira\Desktop\hello.exe
exp(1) = 2.71828
log(1) = 0
log10(100) = 2
isgreater(10,5) = 1
isless(10,5) = 0
isgreaterequal = 1
```

Questions

- Write a C++ program to calculate e^3 using `exp()` and display the result.
- Write a C++ program to calculate the natural logarithm of 50 using `log()` and display the result.
- Write a C++ program to calculate $e^{\log(25)}$ using `exp()` and `log()`.
- Write a C++ program to check whether 45 is greater than 30 using `isgreater()`.
- Write a C++ program to check whether 15 is less than 25 using `isless()`.
- Write a C++ program to check whether 20 is less than or equal to 20 using `islessequal()`.
- Write a C++ program to calculate `exp(2)` and check whether the result is greater than 5.
- Write a C++ program to calculate `log(100)` and check whether the result is less than 5.
- Write a C++ program to calculate `exp(3)` and `log(10)`, and check whether the first result is greater than the second result.
- Write a C++ program to calculate `exp(log(50))` and check whether the result is less than or equal to 50.

Answer : 20.0855, 3.9120, 25,1 (true),1 (true),1 (true),1 (true) ,1 (true), 1 (true),1 (true)

3. <cstring> (character string) OR string.h

Used to work with text data (sequence of characters). It's much easier than C-style character arrays.

Name	Function use	Example
Strlen()	Find length of string	Strlen(hira) = 4
Strcpy()	Copy complete string Syntax : strcpy(destination,source);	a=hira,b; strcpy(b,a) = hira b=hira;
Strncpy()	Copy limited characters Syntax : strncpy(destination,source,number)	A=hira,b; Strncpy(b,a,2) = hi
Strcat()	Concatenate or join strings Syntax : strcat(desti, source)	a= hira , b= kumar; strcat(a,b)=hirakumar
Strncat()	Concatenate limited characters Syntax :strncat(desti,sour,numb)	a=hira,b=kumar; strncat(a,b,3)=hirakum
Strcmp()	Compare two string Syntax : strcmp(1 st stri, 2 nd stri); If 1 st == 2 nd = 0 1 st < 2 nd = any -ve value 1 st > 2 nd = any +ve value	a=hira, b= kumar; strcmp(a,b)= -1 strcmp(b,a) = 1
Strncmp()	Compare first n characters Syntax : strncmp(1 st ,2 nd , 3);	a=hira, b= kumar; strncmp(a,b,3) =
Strchr()	Find first occurrence of character	Char z[]="bananas" Char *p=strchr(z,'a'); p = ananas
Strrchr()	Find last occurrence of character	Char z[]="bananas" Char *p=strrchr(z,'a'); p = as
Strstr()	Find substring and print	
Memcpy()	Copy memory	
Memset()	Set memory	

```
#include<cstring>
using namespace std;
int main()
{
    char a[10] = "hira";
    cout << " Total character length in " << a << " is : " << strlen(a) << endl;

    char b[10];
    strcpy(b,a);
    cout << "Copy other variable : " << b << endl;

    char c[10];
    strncpy(c,b,2);
    cout << "Copy n(number) character in other variable : " << c << endl;

    char e[10]="Kumar";
    strcat(a,e);
    cout << "join character is : " << a << endl;

    strncat(a,e,3);
    cout<< " join limited char : " << a;

    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
Total character length in hira is : 4
Copy other variable : hira
Copy n(number) character in other variable : hi
join character is : hiraKumar
join limited char : hiraKumarKum
```

```
#include<cstring>
using namespace std;
int main()
{
    char a[10] = "hira";
    cout << " Total character length in " << a << " is : " << strlen(a) << endl;

    char b[10];
    strcpy(b,a);
    cout << "Copy other variable : " << b << endl;
    cout << "a & b string are equal ? : " << strcmp(a,b) << endl;

    char c[10]="kumar";
    cout << "a & c are equal ? " << strcmp(c,a) << endl;
    cout << " a & c n number : " << strncmp(a,c,3) << endl;

    char z[10] = "bananas";
    char *p = strchr(z, 'a');
    cout << " p = " << p << endl;
    char *q = strrchr(z,'a');
    cout << " q = " << q << endl;

    char ab[20]="my name is hira";
    char *pp = strstr(ab,"name");
    cout << " pp = " << pp << endl;
}
```

Select C:\Users\hira\Desktop\hello.exe

```
Total character length in hira is : 4
Copy other variable : hira
a & b string are equal ? : 0
a & c are equal ? 1
a & c n number : -1
p = ananas
q = as
pp = name is hira
```

```
#include<iostream>
#include<conio.h>
#include<cstring>
using namespace std;
int main()
{
    char a[10] = "hira";
    cout << " Total character length in " << a << " is : " << strlen(a) << endl;

    char b[10];
    strcpy(b,a);
    cout << "Copy other variable : " << b << endl;
    cout << "a & b string are equal ? : " << strcmp(a,b) << endl;

    char c[10]="kumar";
    cout << "a & c are equal ? " << strcmp(c,a) << endl;
    cout << " a & c n number : " << strncmp(a,c,3) << endl;

    char z[10] = "bananas";
    char *p = strchr(z, 'a');
    cout << " p = " << p << endl;
    char *q = strrchr(z,'a');
    cout << " q = " << q << endl;

    char ab[20]="my name is hira";
```

HIRA KUMAR

```

char *pp = strstr(ab,"name");
cout << " pp = " << pp << endl;

char cc[10];
strncpy(cc,b,2);
cout << "Copy n(number) character in other variable : " << c << endl;

char e[10]="Kumar";
strcat(a,e);
cout << "join character is : " << a << endl;

strncat(a,e,3);
cout<< " join limited char : " << a;

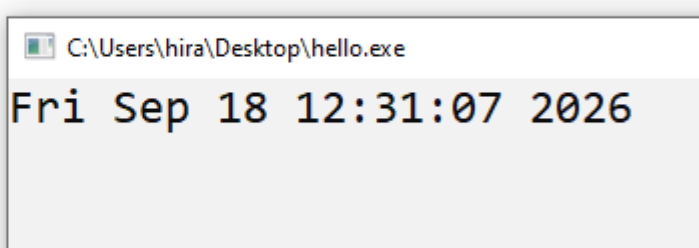
getch();

}

```

4. <ctime>

The <ctime> header file is used to work with date and time in c++.

Function	Use
time()	Returns the current calendar time Syntax : datatype variable; Variable = time(nullptr);
<pre> #include<iostream> #include<conio.h> #include<ctime> using namespace std; int main() { time_t hira; hira = time(nullptr); cout << " Time is (in sec) : " << hira; } </pre> 	
ctime()	ctime() converts a time_t value into a human readable data and time string.
<pre> #include<iostream> #include<conio.h> #include<ctime> using namespace std; int main() { time_t hira; hira = time(nullptr); cout << ctime(&hira); } </pre> 	

Note: Use time(nullptr). The output is: Time in seconds: 1789759352.

This time is continuously calculated in seconds from **1 January 1970** because **1 January 1970, 00:00:00 UTC** was chosen as the standard starting point, called the **Unix Epoch**.

For example

1 Jan 1970 → 0 seconds

1 Jan 1970 + 1 second → 1

1 Jan 1970 + 60 seconds → 60

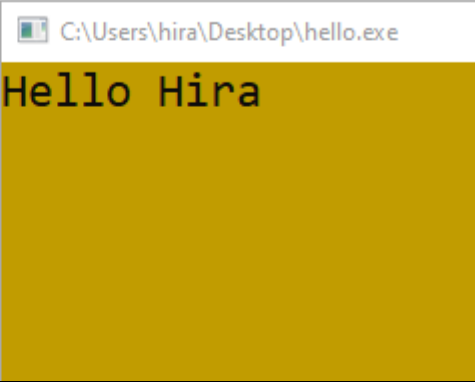
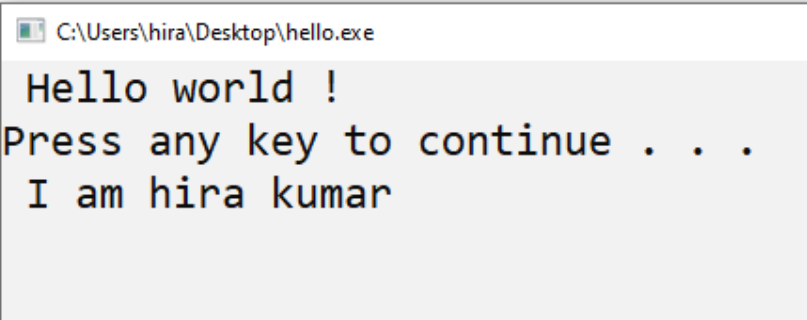
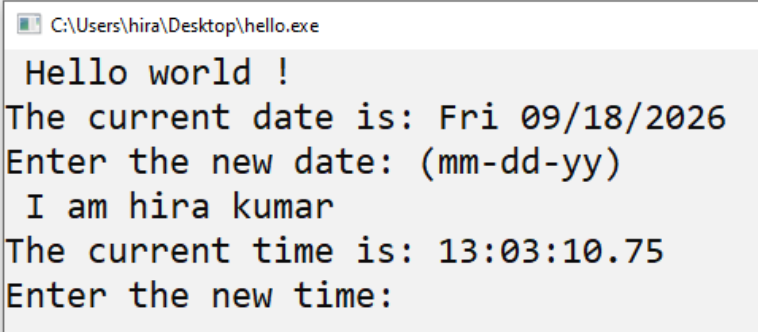
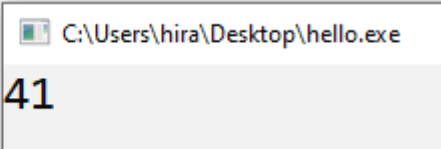
...

2026 → about 1.79 billion seconds

5. <cstdlib>

The <cstdlib> header provides general-purpose utility functions for c++ programs. It contains functions related to random numbers, string to number conversion, memory allocation, program termination, searching/sorting, text and background color change and mathematical utilities.

Function	Use	Example
system()	Change text/background color Syntax : system("color background_color text_color") Common Windows color codes 0 : Black 1 : Blue 2 : Green 3 : Cyan 4 : Red 5 : Magenta 6 : Yellow/Brown 7 : White 8 : Gray 9 : Light Blue 10(A) : Light Green 11(B) : Light Cyan	system("color 70"); or system("color A4");

	12(C) : Light Red 13(D) : Light Magenta 14(E) : Light Yellow 15(F) : Bright White	
<pre>#include<iostream> #include<conio.h> #include<cstdlib> using namespace std; int main() { system("color 60"); cout << "Hello Hira"; }</pre>		
	Clear console screen	system("cls");
	Pause and wait for a key	system("pause");
<pre>#include<iostream> #include<conio.h> #include<cstdlib> using namespace std; int main() { cout << " Hello world !" << endl; system ("pause"); cout << " I am hira kumar " << endl; }</pre>		
	Display/change system date	system("date")
	Display/change system time	system("date")
<pre>#include<iostream> #include<conio.h> #include<cstdlib> using namespace std; int main() { cout << " Hello world !" << endl; system ("date"); cout << " I am hira kumar " << endl; system("time"); }</pre>		
rand()	Generate random number	cout << rand();
	<pre>#include<iostream> #include<conio.h> #include<cstdlib> using namespace std; int main() { cout << rand(); }</pre>	

If you want generate number from b/w 1 to 10 then use formula :
`rand()%10+1`

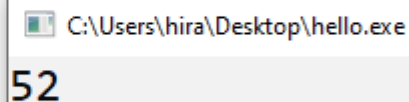
If you want generate number from b/w 1 to 10 then use formula :
`rand()%100+1`

you want your given range :

`rand()%(y-x+1)+x` (here x = starting value , y = last value)

<code>srand()</code>	Initialize the random number generator use <code>time()</code> . So the random sequence changes b/w program runs.
----------------------	---

```
#include<iostream>
#include<conio.h>
#include<cstdlib>
#include<ctime>
using namespace std;
int main()
{
    srand(time(0));
    cout << rand()%100+1;
```

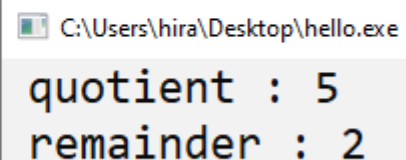


<code>div()</code>	<code>div()</code> performs integer division and returns both : quotient and remainder.
--------------------	---

```
#include<iostream>
#include<conio.h>

using namespace std;
int main()
{
    div_t hira =div(22,4);
    cout << " quotient : " << hira.quot << endl;
    cout << " remainder : " << hira.rem <<endl;

    getch();
```



6. <string>

<string> is a standard C++ header file that provides the `std::string` class and operations for storing and manipulating strings.

```
#include<iostream>
#include<conio.h>
#include<string>
using namespace std;
int main()
{
    string n="hira";
    cout << " My name is " << n ;
    getch();
}
```



Here:

- `#include <string>` → includes the C++ string library.
- `string` → represents a sequence of characters/text.
- `n` → a string variable.
- `"hira"` → string value.

Function	Use	Example
n.length() / n.size()	To find string length	<code>n="hira"; cout<<n.size(); // 4</code>
n.empty()	To check string blank or not	<code>n=""; if(n.empty()); // 1</code>
append()	Add two string	<code>n="hira"; a.append("kumar"); cout<<a; // hira kumar</code>
+	Add two string	<code>string n="hira", n1="kr"; cout<< a+b; // hirakr</code>
push_back()	Adds one character at the end.	<code>string n="hir" n.push_back('a'); cout<<n; // hira</code>
pop_back()	Remove last character	<code>string n="hirag"; n.pop_back(); cout<< n; // hira</code>
substr()	A part of the string coming loose/find	<code>string n="hirahirolal"; cout<<n.substr(4,7); // hiro</code>
find()	Find position of Character/string	<code>cout<< n.find("hiro"); // 4</code>
replace()		<code>n.replace(7,3,"maharaj"); // hirahiromaharaj</code>
erase()		<code>n.erase(4,7); // hiramaharaj</code>
insert()		<code>n.insert(4," g "); // hira g maharaj</code>
stoi()	String to integer	<code>String a="123"; Int n= stoi(a); Cout << n+10; // 133</code>


```

#include<string>
using namespace std;
int main()
{
    string n="hira";
    cout << " My name is " << n << endl;
    cout << "My name is hira" << endl;
    cout << n.length() << endl;
    cout << "-----" << endl;
    if(n.empty())
        cout << " yes (1)" << endl;
    else
        cout<< "no (0)"<< endl;
    cout << "*****" << endl;
    n.append(" hiro ");
    cout << n << endl;
    cout << "-----" << endl;
    string nl="lal";
    cout << n+nl << endl;
    cout << "-----" << endl;
    n.push_back('a');
    cout << n << endl;
    cout << "*****" << endl;
    n.pop_back();
    cout << n << endl;
    cout << "-----" << endl;
    cout << n.substr(4,7) << endl;
    cout << "-----" << endl;
    cout<< n.find("hiro") << endl;

    cout << "*****" << endl;
    n.append("lal");
    cout << n << endl;
    cout << "*****" << endl;
    n.replace(10,3,"maharaj"); //same
    cout << n << endl;
    cout << "*****" << endl;
    n.erase(4,5); // same
    cout << n << endl;
    cout << "*****" << endl;
    n.insert(12," G ");
    cout << n << endl;
    cout << "*****" << endl;
    string a="123";
    int b=stoi(a);
    cout << b+10;
}

```

My name is hira

4

no (0)

hira hiro

hira hiro lal

hira hiro a

hira hiro

hiro

5

```

*****
hira hiro lal
*****
hira hiro maharaj
*****
hira maharaj
*****
hira maharaj G
*****
133

```

7. <algorithm>

Function	Use	Example
max()	Find max value	A=10, b=25; Cout << max(a,b); // 25
min()	Find min value	A=10, b=25; Cout<< min(a,b); //10
swap()	Swap two number	A=10,b=25; Swap(a,b); Cout << a << b; // 25 10
sort()	Sort ascending order of array	Int a[]={10,30,80,20}; sort(a,a+4); for(int x : a) cout << x ; // 10 20 30 80
reverse()	Sort decsending order of array	Int a[]={10,30,80,20};

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		reverse(a,a+4); for(int x:a) cout << x ; // 80 30 20 10
find()	Find number	If(find(a,a+4,30)!= a+4) Cout << "found";
count()	Frequency/repeat of value	Cout<< count(a,a+4,10); //1

```
#include<algorithm>
using namespace std;
int main()
{
int a=10, b=25;
cout << "Max value is " << max(a,b) << endl;
cout << "Min value is " << min(a,b) << endl;
swap(a,b);
cout << "swap value is a = " << a << ", b = " << b << endl;
int hira[]={10,30,80,20};
sort(hira,hira+4);
for(int z : hira)
cout << z << " " << endl;
reverse(hira,hira+4);
for(int z1 : hira)
cout << z1 << " ";
cout << count (hira,hira+4,10) << endl;
if(find(hira,hira+4,30)!=hira+4)
cout<<"found";
getch();
}
```

```
Max value is 25
Min value is 10
swap value is a = 25, b = 10
10
20
30
80
80 30 20 10 1
found
```

8. <iomanip>

9. <cstdio> → use printf(),scanf() in c++ language use this header file

10. <vector>

11. <fstream>

12. <cctype>

13.so many